

RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Academic Year: 2019 - 2020 (Even Semester)
INNOVATIVE PRACTICES

Degree, Semester & Branch: IV Semester B.E. CSE Course

Code & Title: CS8491 Computer Architecture

Name of the Faculty member: Mr. A.S.Prabaharan

Active Learning Strategy: Role Play

Topic: Pipeline Processing and Hazards

Date: 14.02.2020

Role play:

Role play encourages participation among passive learners, adds dynamism to the classroom and promotes the retention of material.

Benefit of the Role play:

- Students immediately apply content in a relevant, real world context.
- Students can transcend and think beyond the confines of the classroom setting.
- Students see the relevance of the content for handling real world situations.
- The instructor and students receive immediate feedback with regard to student understanding of the content.
- Students engage in higher order thinking and learn content in a deeper way.
- Instructors can create useful scenarios when setting the parameters of the role play when real scenarios or contexts might not be readily available.

Course Outcome:

CO3: Describe the purposes of data path and control path, pipelined execution of instructions and its hazards.

Goal:

The students will be able to implement Pipeline processing and its hazards.

Reason for choosing the particular topic (Method):

Students are asked to “act out” so they get a better idea of the concepts and theories being discussed. Role play helps the students to visualize the functioning of pipeline processing and its hazards. In addition, role play is used to empower, engage, and stimulate a classroom by putting students at the centre of the learning process.

How we implemented Role play:

- The faculty has discussed the concept of pipeline processing its hazards on the previous day and asked willingness from the students to role plays the concepts on the next day.
- The students formed groups and prepared for the role play.
- Students have taken the role of pipeline processor stages and instruction execution.
- The students acted as sequential and pipeline processor.

Learning Outcomes:

The students were able to demonstrate Pipeline Processing and its hazards.



Reflective Critique:

Pre-implementation Reflection:

- Some students were eagerly to participate which necessary for the execution of the role is playing activity.
- More number of students was involved in the activity.

Post Implementation reflection:

- Students were able to identify and apply the working of pipeline processor.
- All the students enjoyed the activity.
- Students' feedback reflected that they have understood the concept.

Learning Outcomes/ Program Outcomes	PO1	PO2	PO3	PO9	PO10	PO12
LO: The students will be able to implement Pipeline processing and its hazards.	3	3	1	2	1	2

References: <https://serc.carleton.edu/introgeo/roleplaying/whatis.html>
https://www.ritrjpm.ac.in/images/computer-science/10_CS8591_Rolepaly.pdf
<https://www.ritrjpm.ac.in/images/computer-science/CS8493-OS-Roleplay.pdf>

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